

Mohammad

Mechanical Construction, Sensor Mounting, Welding, and Prototype Accessories

Mohammad was mainly responsible for the mechanical construction and physical assembly of the Smart Weather Station. His work focused on securing the components, preparing the prototype structure, and ensuring that the sensors and mechanical parts could operate correctly during testing and demonstration.

He welded, soldered, or mechanically secured the required sensor connections and helped reinforce components that could not remain stable using temporary breadboard wiring alone. He worked on fixing the load cell and HX711 platform so that weight could be applied correctly without excessive movement. Proper load-cell mounting was essential because vibration, platform contact, and off-center loading can affect the measured value.



Mohammad was also responsible for installing and securing the water-flow sensor. This included arranging the tubing, supporting the sensor body, checking the water-flow direction, and reducing leakage or movement during testing. He helped prepare the collection arrangement used during flow testing and calibration.

In addition, Mohammad designed and constructed the physical accessories used to present the project. These included the model house, the clothesline structure, the servo mechanism, sensor supports, platforms, covers, and other decorative or functional parts. These accessories transformed the connected components into a coherent Smart Weather Station prototype and made the automatic clothesline feature easier to demonstrate.



He also assisted during final assembly and system-level testing by adjusting component positions, repairing mechanical problems, supporting cables, and ensuring that the prototype remained stable during operation and video recording.



Personal reflection:

This project helped me understand how mechanical design affects sensor performance. I learned that a correct circuit and program are not enough if the sensors are mounted poorly or the physical structure is unstable. The load cell required a rigid and correctly positioned platform, while the flow sensor required secure tubing and correct orientation. I also gained experience constructing a prototype that was both functional and suitable for presentation.